

Chapter 1



INTRODUCTION TO DATABASES TRANSPARENCIES

Chapter 1 - Objectives



- **Some common uses of database systems.**
- **Characteristics of file-based systems.**
- **Problems with file-based approach.**
- **Meaning of the term database.**
- **Meaning of the term Database Management System (DBMS).**

Chapter 1 - Objectives



- **Typical functions of a DBMS.**
- **Major components of the DBMS environment.**
- **Personnel involved in the DBMS environment.**
- **History of the development of DBMSs.**
- **Advantages and disadvantages of DBMSs.**

Examples of Database Applications



- **Purchases from the supermarket**
- **Purchases using your credit card**
- **Booking a holiday at the travel agents**
- **Using the local library**
- **Taking out insurance**
- **Renting a video**
- **Using the Internet**
- **Studying at university**

File-Based Systems



- **Collection of application programs that perform services for the end users (e.g. reports).**
- **Each program defines and manages its own data.**

File-Based Processing

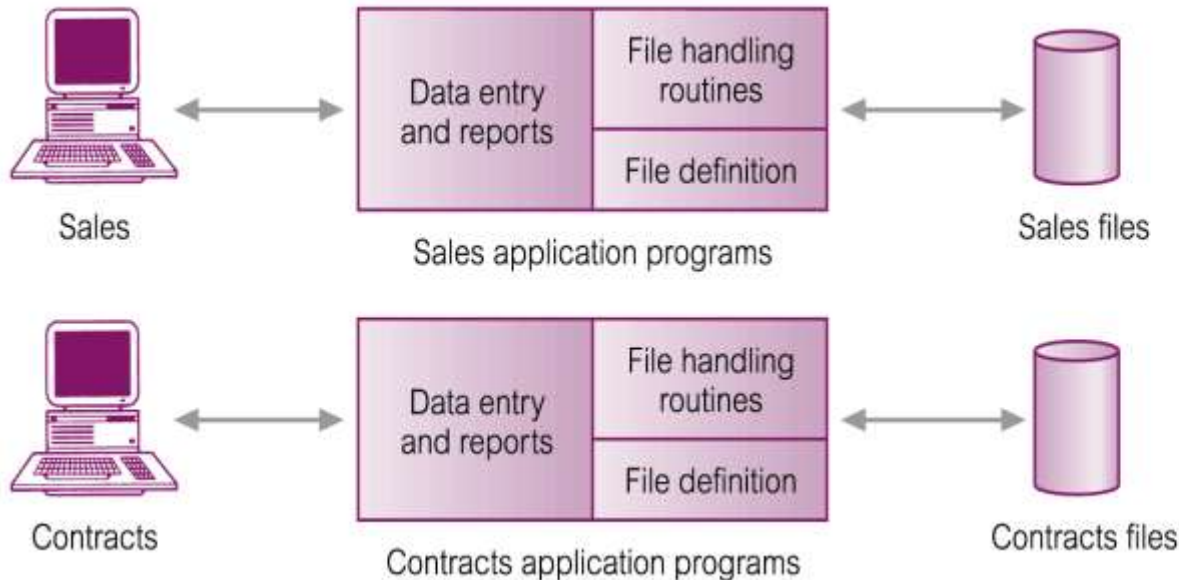


Figure 1.5

File-based processing.

Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Limitations of File-Based Approach



- **Separation and isolation of data**
 - Each program maintains its own set of data.
 - Users of one program may be unaware of potentially useful data held by other programs.

- **Duplication of data**
 - Same data is held by different programs.
 - Wasted space and potentially different values and/or different formats for the same item.

Limitations of File-Based Approach



- **Data dependence**
 - File structure is defined in the program code.
- **Incompatible file formats**
 - Programs are written in different languages, and so cannot easily access each other's files.
- **Fixed Queries/Proliferation of application programs**
 - Programs are written to satisfy particular functions.
 - Any new requirement needs a new program.

Database Approach



- **Arose because:**
 - Definition of data was embedded in application programs, rather than being stored separately and independently.
 - No control over access and manipulation of data beyond that imposed by application programs.
- **Result:**
 - the database and Database Management System (DBMS).

Database



- **Shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.**
- **System catalog (metadata) provides description of data to enable program–data independence.**
- **Logically related data comprises entities, attributes, and relationships of an organization's information.**

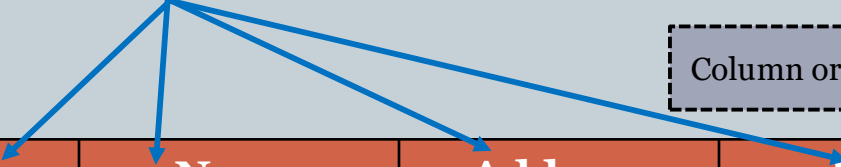
Database



Database Name: FIKStudent

Table Name: DTMM

Column or Field or Attributes



ID	Name	Address	Fees
1	Liyana	Kuala Terengganu	1200
2	Alif	Kuala Selangor	600
3	Intan	Kuala Lumpur	1200

File-Based vs DBMS



	advantages	disadvantages
FMS	• Simpler to use • Less expensive	• Typically no multi-user access • Limited to smaller databases • Limited functionality • Decentralization of data • Redundancy and integrity issues
DBMS	• Greater flexibility • Greater processing power • Ensures data integrity • Supports simultaneous access • Provides backup and recovery controls • Advanced security	• Difficult to learn • Packaged separately from the OS • Slower processing speeds • Requires skilled administrators • Expensive

Database Management System (DBMS)



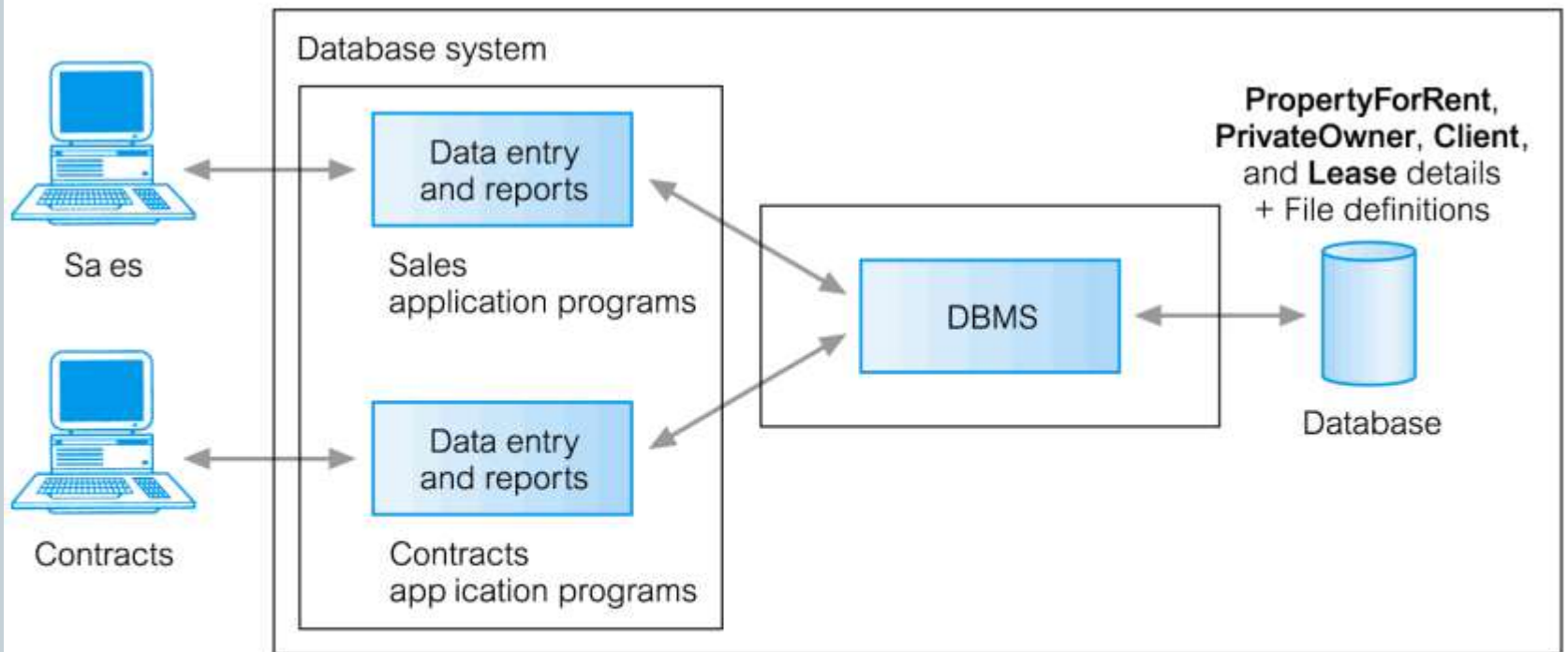
- A software system that enables users to define, create, maintain, and control access to the database.
- (Database) application program: a computer program that interacts with database by issuing an appropriate request (SQL statement) to the DBMS.

```
-- get employees who joined company in 2000
SELECT clause { SELECT first_name
FROM clause { FROM employees
WHERE clause { WHERE YEAR(hire_date) = 2000
```

Comment

Predicate

Database Management System (DBMS)



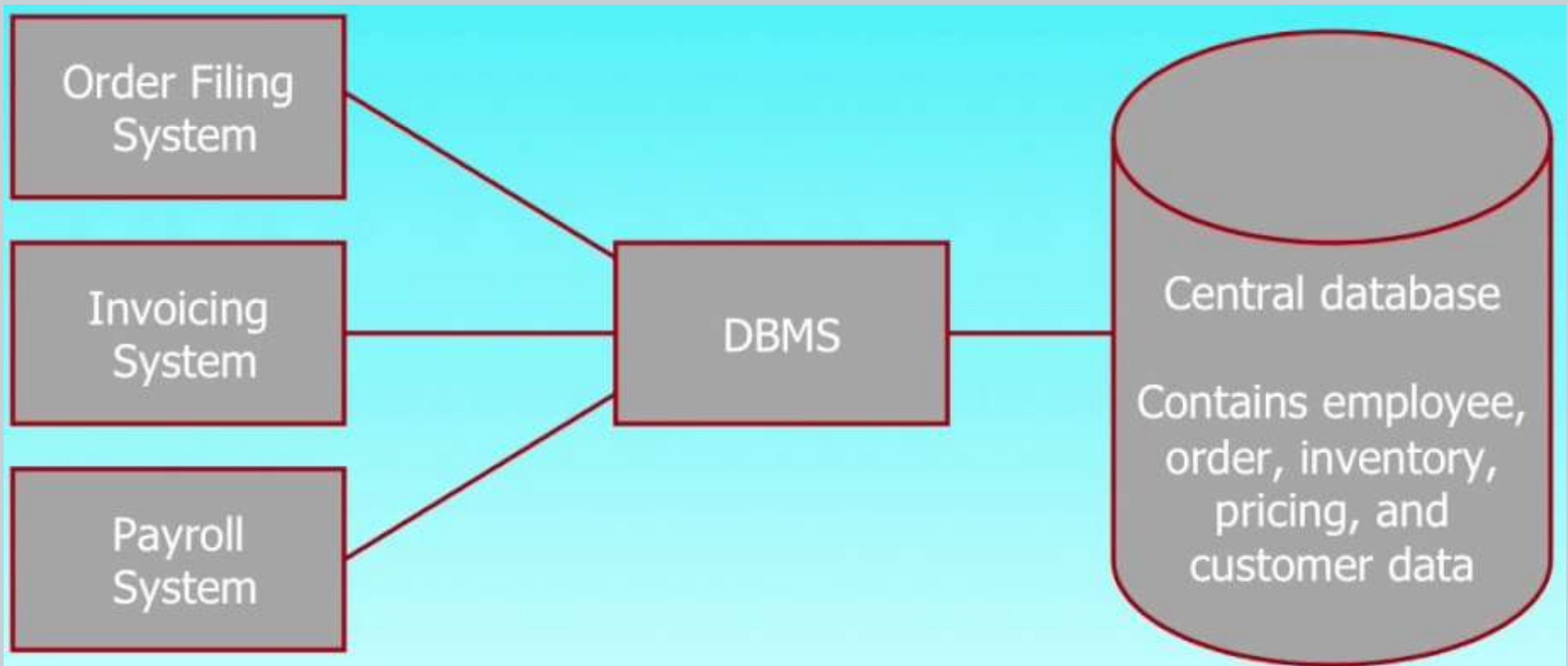
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Database Management System (DBMS)



Database Approach



- **Data definition language (DDL).**
 - Permits specification of data types, structures and any data constraints.
 - All specifications are stored in the database.
- **Data manipulation language (DML).**
 - General enquiry facility (query language) of the data.

Database Approach



DDL	DML
It is Data Definition Language	It is Data Manipulation Language
These are used to define data structure	It is used to manipulate the existing databases.
It is used to define database structure or schema	It is used for managing data within schema objects
Commands are: CREATE, ALTER, DROP, TRUNCATE, RENAME	Commands are: SELECT, INSERT, DELETE, UPDATE, MERGE, CALL
It works on whole table	It works on one or more rows
It do not have a where clause to filter	It have where clause to filter records
Changes done by DDL commands cannot be rolled back	Changes can be rolled back
It is not further classified.	It is further classified as procedural and <u>non procedural</u> DML's
Example:- drop table <u>tablename</u> ;	Select * from employee

Database Approach



- **Controlled access to database may include:**
 - a security system
 - an integrity system
 - a concurrency control system
 - a recovery control system
 - a user-accessible catalog.

Views



- **Allows each user to have his or her own view of the database.**
- **A view is essentially some subset of the database.**

Database: Agranibank.mdb

Table: Customer

Name	Account	Address
Karim	101	H-314
Rahim	102	H-620
Kamal	103	H-540

Table: Balance

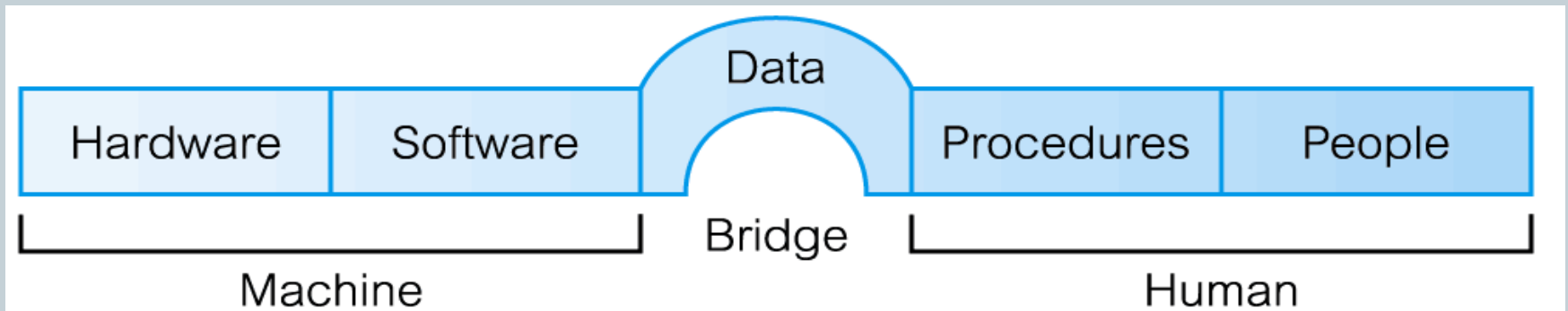
Account	Balance
101	7000
102	5000
103	8000

Views - Benefits



- **Reduce complexity**
- **Provide a level of security**
- **Provide a mechanism to customize the appearance of the database**
- **Present a consistent, unchanging picture of the structure of the database, even if the underlying database is changed**

Components of DBMS Environment



Components of DBMS Environment



- **Hardware**
 - Can range from a PC to a network of computers.
- **Software**
 - DBMS, operating system, network software (if necessary) and also the application programs.
- **Data**
 - Used by the organization and a description of this data called the schema.
- **Procedures**
 - Instructions and rules that should be applied to the design and use of the database and DBMS.
- **People**

Roles in the Database Environment



- **Data Administrator (DA)**
- **Database Administrator (DBA)**
- **Database Designers (Logical and Physical)**
- **Application Programmers**
- **End Users (naive and sophisticated)**

History of Database Systems



- **First-generation**
 - Hierarchical and Network
- **Second generation**
 - Relational
- **Third generation**
 - Object-Relational
 - Object-Oriented

Advantages of DBMSs



- **Control of data redundancy**
- **Data consistency**
- **More information from the same amount of data**
- **Sharing of data**
- **Improved data integrity**
- **Improved security**
- **Enforcement of standards**
- **Economy of scale**

Advantages of DBMSs



- **Balance conflicting requirements**
- **Improved data accessibility and responsiveness**
- **Increased productivity**
- **Improved maintenance through data independence**
- **Increased concurrency**
- **Improved backup and recovery services**

Disadvantages of DBMSs



- **Complexity**
- **Size**
- **Cost of DBMS**
- **Additional hardware costs**
- **Cost of conversion**
- **Performance**
- **Higher impact of a failure**



END OF CHAPTER 1..